
Trust-Calibrated Multi-Agent Scientific Deliberation for Mitigating Sycophantic Consensus in LLM Reasoning

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Abstract

Large language models have done surprisingly well at answering many scientific questions. However, large language models also continue to "hallucinate" and reason incorrectly. One potential solution to these problems is multi-agent debate. Although a correct minority may exist, a confident but incorrect majority can drag this correct minority agent down and lead the group to select the wrong answer. This is known as the "sycophantic consensus." Currently, there is no mechanism within any debate system to check an individual agent's assertions against outside scientific evidence before allowing them to gain influence in the debate. Instead of using a simple vote-weighted approach, we intend to assign a trust score to every participating agent based on how much scientific evidence supports their responses. Every time agents respond during a round, the system breaks down their responses into small units of factually verifiable statements ("claims"). After breaking down the response, the system locates all relevant scientific literature that could support or contradict each claim. Because unsupported claims eventually lose influence and because the trust scores are kept in bounds, the probability increases that a correct minority will survive long enough to ultimately win. Our proposed framework produced fewer hallucinated answers than competing frameworks because all surviving claims are linked back to their originating passages of scientific evidence. To test this hypothesis, we compare our proposed framework to majority voting and other baseline systems. Once all models have finished responding to a given question, the system returns both the selected model and supporting citations for that model. The performance of our proposed framework is evaluated via four metrics: answer accuracy, consensus collapse, minority preservation, and evidence calibration.

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Our FYDP instructor Dr. Ohidujjaman set the deadlines and help with the report. Dr. Ohidujjaman's timely responses allowed the team to maintain a consistent pace throughout the trimesters.

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Publication List

The publications connected to this research are listed below. One journal article is currently under review, and further submissions are planned once the evaluation results are ready.

Journal Articles

1. Trust-Calibrated Multi-Agent Scientific Deliberation for Mitigating Sycophantic Consensus in LLM Reasoning (In preparation)

Conference Papers

- 1.

Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

- 1.

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Chapter 1

Introduction

This chapter sets the scene for the whole report. Background and problem come first, followed by the motivation, the objectives, a short account of the methodology, the expected outcome, and the structure of the remaining chapters.

1.1 Project Overview

Large language models today have been shown to be capable of handling a wide variety of reasoning tasks such as answering scientific questions, serving as tutors or providing decision support. A large language model can still be confidently wrong. It may state an incorrect answer in fluent and assertive language that hides the error. To address this, the use of multi-agent debate has been suggested [1]. In a debate, multiple agents respond to the same question, and they review the rationales provided by other agents over several iterations. After reviewing all rationales, the agents select a single answer [1]. Chain-of-thought prompting gives each agent a way to expose its intermediate steps [2], which makes it easier for others to evaluate the rationale behind the agent’s position. A significant challenge exists at the last stage of the process. This involves determining how best to combine separate answers produced by individual agents. The majority of current debate-based systems resolve disputes through a simple voting mechanism [3]. In these mechanisms, each agent receives equal weight (i.e., each vote counts equally), and the answer receiving the highest number of votes ultimately prevails [4]. Such a system evaluates correctness solely on the basis of the relative frequency of occurrence of an answer among agents. However, in scientific inquiry, the quality of an argument is not determined by the number of times it is repeated; instead, arguments are considered valid when supported by credible evidence. The study presents a trust-calibrated multi-agent deliberation framework that modifies the influence agents have in a debate based on how well their statements align with evidence obtained via retrieval from external sources. Therefore, when evaluating the validity of a statement in the context of a multi-agent deliberative environment, the study suggests using a measure of evidence-grounded trust rather than a raw count of heads.

1.2 Motivation

Majority voting can fail in a way that is easy to miss. A debate may look like it reached agreement while it has quietly dropped a correct answer held by one agent. Recent studies of multi-agent debate report that agents often copy or switch answers under social pressure instead of reasoning on their own. The correct answer is present in the discussion but ignored in more than 20 percent of wrong-answer cases [5]. Work on model behaviour points to a related tendency, where models trained to be agreeable grow more sycophantic and less accurate [6]. A confident majority pushes a correct minority agent to give up its answer, and majority voting then does more than pick the wrong result. It hides the failure behind an appearance of consensus. This matters most for scientific questions, where a wrong but confident consensus can mislead a user into trusting an answer that deserved a closer check. Each existing method attacks only part of the problem. iMAD decides when a debate is worth running at all [7]. DebUnc scales agent influence by self-reported confidence [8]. Mixture-of-Agents folds all outputs together through static aggregation [9]. None of these ties an agent’s influence to whether its claims survive a check against external evidence while the debate is still running. That connection is exactly what this study adds.

1.3 Objectives

The main objective is to reduce sycophantic consensus in multi-agent LLM debate. The study replaces plain majority voting with trust that is grounded in evidence. To do this, the work first needs a clear picture of the failure. A confident but wrong majority can push a correct minority agent to change its answer [5, 10]. Next, it needs a bounded trust score. The score rises when external evidence supports an agent’s claims and falls when the evidence contradicts them. The update happens before the final answer is fixed. The study also tests the method against majority voting and other debate and aggregation baselines [3, 4]. Performance is measured with four rates: answer accuracy, consensus-collapse, minority-preservation, and evidence-calibration.

1.4 Methodology

The study uses an experimental design. A baseline debate system is built first. Several heterogeneous LLM agents answer the same scientific question and revise their positions after reading each other’s reasoning across rounds [1]. The baseline keeps majority voting. This makes the usual failure mode visible and lets it be measured [4]. Next, each agent’s reasoning is split into smaller factual claims. These claims are checked against scientific evidence from external sources, such as academic papers and reference databases [11]. Contradictions between a claim and the retrieved evidence are detected and scored [12]. The idea that external tools can verify and correct model output supports this step [13]. Verdicts then feed into trust updates. This follows work that uses per-agent confidence

to steer debate [8]. A supported claim raises the agent’s trust, and a contradicted claim lowers it. Uncertain claims are handled in a conservative way so the score does not jump on weak evidence. The trust score stays bounded, so no agent gains unlimited influence or drops out completely. The final answer comes from trust-weighted aggregation, not a simple count. The framework is tested on scientific reasoning datasets such as GPQA and MMLU-Pro. Evaluation checks whether the mechanism lowers consensus collapse and whether correct minority answers survive. It also checks whether overall accuracy holds, all without any fine-tuning. Figure 1.1 shows the end-to-end flow.

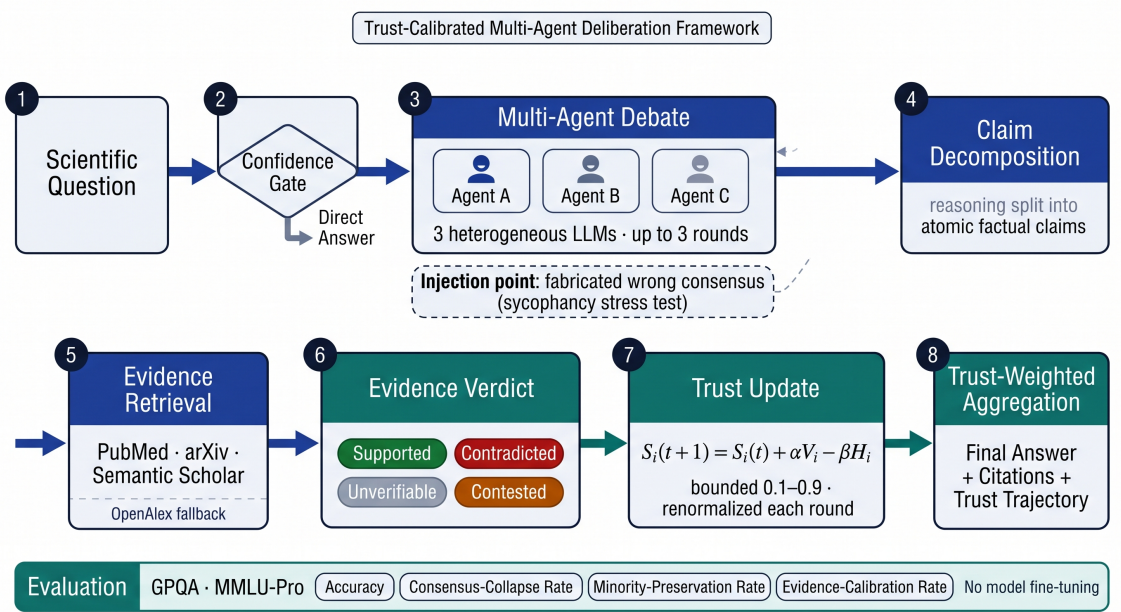


Figure 1.1: End-to-end flow of the trust-calibrated multi-agent deliberation framework: debate, claim-level evidence verification, per-round trust updates, and trust-weighted aggregation.

1.5 Project Outcome

The main outcome is a reproducible framework for trust-calibrated scientific deliberation with multiple agents. Agent influence is updated from retrieved evidence while the debate runs. The final answer then rests on evidence-weighted trust, not a majority vote. The study also produces an evaluation protocol. The protocol detects sycophantic consensus [5] and tracks how often correct minority answers survive [10]. It also measures the method against established baselines. Dynamic trust calibration should make debate more dependable on scientific questions. A wrong answer can still remain. The decision is still more transparent because the method records which agents the evidence supported and which it contradicted.

1.6 Organization of the Report

The rest of the report moves from background to results. Chapter 2 covers multi-agent debate and sycophancy. It also defines the gap this study addresses. The requirements, the methodology, the system design, and the project plan sit in Chapter 3.

Chapter 4 holds the implementation setup and the testing procedure. It also presents the results with discussion. Standards, constraints, sustainability, and the complex engineering problem analysis fill Chapter 5. Chapter 6 closes with a summary and limitations. It also outlines directions for future work.

Chapter 2

Background

The chapter introduces the concepts needed to understand the system described in the report and reviews research on multi-agent debate, sycophancy, and agent influence. It then explains the gap addressed by the study.

2.1 Preliminaries

There are several techniques for developing "trust" in an Artificial Intelligence (AI) system. Some of these approaches depend upon a consensus of answers generated by multiple AI systems. Other techniques depend upon finding relevant scientific information and assessing the accuracy of the information found. Chain-of-Thought prompts assist in illustrating the reasoning process that a model uses to arrive at its conclusion, so that we may better understand how an answer was produced and why an error was made during an instance of poor performance [2, 14]. A Multi-Agent Debate Framework provides a methodology where each agent evaluates the arguments presented by the other agents. Each agent presents their proposed solutions and includes support for those proposed solutions. All agents evaluate the proposed solutions and supporting rationales provided by the other agents. Following evaluation and revision of their respective positions and solutions, a final proposed solution is selected [1]. One advantage of allowing each agent to independently evaluate a problem is that if each agent discovers different faults in the system, then collectively the entire team may identify potential problems with the system. Additionally, a team member can utilize another team member's rationalization to criticize aspects of their own rationalization. However, regardless of how well the individual components of the solution are developed, the combination of results (i.e., the methodology of combining results) is equally as important as the results themselves. Most commonly, results are combined utilizing a majority vote. In a majority vote, every member has an equal say, and the solution supported by the highest number of members is therefore selected. However, majority voting does not necessarily eliminate errors; rather, it can strengthen repeated errors [3]. Sycophantic Consensus occurs when an agent agrees with the majority position or changes a correct position in favor of a more strongly supported

but incorrect position [5]. There are two types of confidence associated with voting: Self-Confidence in one’s position and Token Distribution Confidence. Neither type of confidence verifies the truth claims contained within a position [8]. DebUnc identifies this weakness by comparing uncertainty metrics produced by the deployed system with a Diagnostic Ground Truth Metric that already contains the known answer [8]. Developing trust in an AI system therefore requires obtaining and evaluating scientifically relevant data and evidence.

2.2 Literature Review

The review categorizes the reviewed literature by five categories: debate construction, external verification, sycophancy, uncertainty, and debate evaluation. This study primarily compares methods for determining when to generate a response (i.e., methods for deciding on debate), methods for changing how an agent interacts with another agent (i.e., methods for changing agent interaction), and the method for changing what influences an agent’s decision based upon information retrieved from outside the system (i.e., changing agent influence). Recent surveys are provided in [15] as they outline the design space of multi-agent systems utilizing LLMs. In order to develop generated responses that are more reliable, research has shifted focus toward structured reasoning, multi-agent collaboration, and external validation of responses generated by LLMs [14]. Multi-Agent Debate is an example of this new area of research because Multi-Agent Debate allows for other agents to evaluate and potentially challenge each agent’s reasoning before that agent produces their final output [1]. The next section outlines some of the most relevant current works in this emerging area and illustrates how the proposed architecture addresses an existing research gap.

2.2.1 Similar Applications

It should be noted that this system is very different from a standard web/mobility application. In particular, the system we are building is an experimental layer designed to enable deliberative processes and act as an integration layer for scientific question answering systems, research assistants, evidence based decision support tools and/or other applications requiring reliable reasoning. As opposed to the traditional approach where one model solution is generated, the systems listed above generate multiple candidate solutions. Additionally, the systems listed above allow for interaction among candidates. Finally, each system aggregates results from the candidates to generate a final answer for users. Du et al. showed that structured debate among several agents increases factuality and reasoning compared to single-agent approaches [1]. Wang et al. showed that layered aggregation can result in higher overall model capability when combining outputs from several models without supporting explicit debate [9]. Multi-agents systems differ primarily regarding how output is combined. Some systems utilize structured debate as a

means to enhance reasoning between agents [1], while others use hierarchical pipelines to aggregate/average the output from each pipeline (the individual agents) [9]. Zhou et al. proposed a self-organizing system for tasks to determine whether a specialized reasoner produces the response versus being in line with a pre-defined workflow [16]. Additionally, Kushwaha & Madhavan further developed an augmented retrieval framework that identifies contradictions found within retrieved documents and ranks credibility of the source prior to producing the response [12]. Both are examples of how distributed reasoning (across multiple specialized agents) results in increased robustness/accuracy in factually correct responses. What matters is on what scale you construct dialogue. Ueda et al. analyzed how to construct multi-agent LLM dialogues to facilitate research ideation [17]; however, Wang et al. studied if/how large-scale multi-agent LLMs could provide assistance with scientific code generation [18]. Their aggregation method does not change during dialogue. Similar to a single LLM, the proposed framework has the same reliability goals for answers; yet allows for various levels of influence for each agent throughout the discussion process. After each round of dialogue, evidence verification results are generated; and after each round, the trust scores for all agents are updated based on claim-level scientific evidence supporting their claims. Therefore, an unsupported majority consensus would hold less value than a well-supported minority view.

2.2.2 Related Research

The work of Wei et al. (2022) shows that “Chain-of-Thought Prompting” improves the ability of a large language model to reason [2]. The work of Du et al. (2024) extended the work of Wei et al. (2022) by incorporating multi-agent debate in addition to chain-of-thought prompting, in which multiple agents give answers to a question, look at how other agents have reasoned when answering the question, and then adjust their reasoning [1]. In contrast to both previous approaches for enhancing the performance of large language models, CRITIC [13] relies on outside tools to validate the answers provided and enhance them. It therefore permits validation of assertions contained in each response; however, there is no mechanism to compare additional models working on the same question simultaneously. Instead, Mixture-of-Agents develops numerous independent agent-proposer models that produce answers to identical questions and layers the responses into one output, indicating what several large language models can do together; however, the layering of the individual agent responses is fixed and contains no evidence-based trust values [9]. iMAD addresses cost rather than correctness [7]. Fan, Yoon, and Ji identified 41 properties of a structurally defined self-critic and constructed a classifier that determines whether or not to initiate a debate based on those properties. They were able to attain token reductions of up to 92% relative to GroupDebate and increases in accuracy of up to 13.5% over a single-agent baseline. Agent influence stayed constant throughout the debate. Pitre, Ramakrishnan, and Wang investigated sycophancy in debates [5]. The results of their investigation indicate that the correct answer existed in more than 20% of cases where the

agents responded incorrectly. CONSENSAGENT identifies stalling and answer copying in a debate. When either condition is identified, CONSENSAGENT alters the task prompt before initiating the next stage of the debate. The results of the study demonstrate that measured sycophancy decreased as a result; however, the total trust score is still dependent upon self-reported confidence and consistency. DebUnc permits agents to express uncertainty among themselves [8]. However, agents express uncertainty via prompts or attention scaling. The deployed measures provided very limited benefit. A diagnostic ground-truth signal is available for comparison purposes. The diagnostic ground-truth signal has access to information concerning the correct answer and has greater potential than the deployed measures. The difference between the deployed measures and the diagnostic ground-truth signal indicates that the major barrier is the quality of the trust signal. Kushwaha and Madhavan created an 8-agent MAS-RAG system intended to identify contradictions between retrieved documents and rank evidence based upon credibility [12]. Although source credibility was evaluated by humans prior to deliberation, it was never changed during deliberation. There have also been numerous studies evaluating the social aspects of debate. The authors of Cau et al. identified how selective persuasion could cause an increase in opinion convergence [19]. Yeow et al. studied decision architectures and found that feedback between agents can spread early errors [20]. Wang et al. demonstrated how a good example given in context within one large language model (LLM) may perform better than multi-agent dialogue in certain situations. As noted in [3], judge error and peer conformity were identified as potential drawbacks. The authors of Choi, Zhu, and Li separated communication used in debate from the act of voting and found that majority voting explained a substantial portion of the normal performance gains [4]. FREE-MAD extended previous research by eliminating the need for final agreement and utilizing anti-conformity prompts and trajectory scores [21]. ReSo utilizes dynamic task graphs with reward-based processes for choosing agents [16]. Even though past research has explored optimizing coordination and the choice of agents, none of the previously conducted studies have assessed scientific claims prior to altering the influence of an agent. Sycophancy also occurs at the model level. The results of training models to be perceived as “warm” and “agreeable,” as demonstrated by Ibrahim et al., produced lower levels of factual accuracy and higher levels of sycophancy [6]. Chen et al. utilized self-augmented preference alignment from a training perspective to address the issue [22]. Process-level diagnostics for engagement, responsiveness, influence asymmetry, balance, stability, and agent utility developed by Pitre et al. would be helpful when ground truth is unavailable [23]. Final agreement is insufficient to measure a debate, as stated by Pitre et al. [23].

2.3 Gap Analysis

There are still three open issues with regard to Multi-Agent Debate Systems (MADS) from our review of the current literature. The first issue is whether the extra computational burden associated with a MADS can be justified. Fan et al. have used iMAD for

this purpose – to predict whether a debate is needed prior to the start of a debate [7]. The second issue is how to detect unsupported agreement caused by conformity or sycophancy. CONSENSAGENT addresses this problem through prompt optimization during the debate [5]. The third issue is how to determine which agents should receive greater influence when they disagree with one another. DebUnc has a weighting mechanism to deal with uncertainty as it relates to the debate process; however, the trust mechanisms of this weighting rely on each individual agent’s internal assessment of uncertainty and therefore do not use external validation or independently verifiable evidence [8]. As previously mentioned, even when multiple agents agree with respect to a particular piece of information, that information may be incorrect. For example, Cau et al. showed that selective persuasion can affect the level of consensus reached by different agents [19], while Wang et al. showed that peer conformity can limit the effectiveness of multi-agent debate [3]. Existing retrieval-based systems verify the correctness of retrieved information or improve the aggregation process, but they do not dynamically update an agent’s influence based upon verified scientific evidence during a debate [13]. Similarly, although dynamic coordination mechanisms help solve the issue of selecting which agents participate in the debate, they do not provide a mechanism for increasing or decreasing an agent’s influence based on evidence of its behavior during the debate [16]. The existing research therefore does not currently contain a method by which an agent’s influence can be continuously updated using independently verified evidence of its behavior in real time. The proposed framework addresses this remaining gap by decomposing each agent’s responses into individual factual claims and identifying scientific passages that relate to those claims. Each claim is then classified as either supported, contradicted, or unverifiable. The classification of each claim serves as input to an update function for a bounded trust score. As a result, well-supported minority agents can continue to have influence during a debate, while an unsupported majority consensus is given less weight. To evaluate the proposed framework, we will use a number of metrics, including Consensus-Collapse Rate (CCR), Minority-Preservation Rate (MPR), Evidence-Calibration Rate (ECR), and Answer Accuracy (AA), alongside conventional answer accuracy measures.

2.4 Summary

The chapter introduced LLM reasoning, chain-of-thought prompting, multi-agent debate, and sycophantic consensus. The reviewed studies cover debate efficiency, prompt correction, uncertainty weighting, external verification, process evaluation, and agent coordination. They do not continuously update agent influence from claim-level scientific evidence. The next chapter uses this gap to define the system requirements, detailed methodology, and project plan.

Chapter 3

Project Design

Chapter 3 defines the system requirements and lays out the methodology, the design decisions, and the project plan with the task allocation. The first section records the functional and nonfunctional requirements together with the context diagram, the level-one data flow diagram, and the interface design.

3.1 Requirement Analysis

Requirement analysis fixes what the framework must do before any code is written. The system is an experimental deliberation layer rather than a conventional application, so the requirements are framed around the research pipeline. A question enters, agents debate it over a fixed number of rounds, the claims are checked against external evidence, and a trust-weighted answer comes out. A multi-agent debate approach is used here since prior research indicates it yields more factual responses than relying on a single model [1]. Furthermore, we incorporated a trust mechanism to prevent situations where a confident but incorrect majority forces a correct minority agent to change its answer [5]. The below sections will provide detail on the individual components of the System Requirements for this project including Boundaries, Data Flows and the User Interface for the Framework.

3.1.1 Functional and Nonfunctional Requirements

The Functional Requirements outline the primary functions that the framework provides. All input questions (from both the User and/or the Evaluation Harness) are processed in complete ignorance of their actual "ground-truth" as it relates to the Agents. In order to be efficient, all input questions do not require a full debate. Therefore, an initial Confidence Gate determines whether an input question should receive a Direct Response or be sent to the Main Pipeline for processing. The Orchestration Process controls 3 different Heterogeneous LLM Agents, each revising at most 3 times per Agent (via State Machine), as determined by 3 retry limits.

Each Agent's Reasoning is segmented into Atomic Factual Claims and carefully tagged to its Source. If this structured tagging process fails, a fallback extraction pass is triggered

to handle the unparsed text. Retrieval gives each agent its own database: PubMed for the first, arXiv for the second, Semantic Scholar for the third. OpenAlex stands in when a source is down, and a cross-encoder reranks the passages that come back. Each claim then lands on one of four verdicts: supported, contradicted, unverifiable, or contested.

After every round the trust score is updated as $S_i(t+1) = S_i(t) + \alpha V_i - \beta H_i$. A softmax follows, the values are clamped between 0.1 and 0.9, and the vector is renormalized.

Those two bounds matter. Without them an agent could be silenced outright, or could grow to dominate the debate. An injection point lets the harness insert a fabricated wrong consensus between rounds. That is how sycophantic collapse gets measured under controlled pressure. At the end, weighted aggregation over trust and position picks the final answer, and the majority vote plays no part in it. The results package contains the answers, the citations supporting those answers, the full trust history, and the reasoning trace for each agent.

The evaluation harness will compute an estimate of the answer’s correctness along with a measure of how well the framework has collapsed into a consensus, preserved minority opinions, and calibrated its use of evidence. It will evaluate this framework relative to baseline methods based upon majority voting and MoA [9], iMAD [7], and DebUnc [8]. There are several built-in recovery routes in case there is some kind of failure within the system. For example, if a model’s output cannot be processed then it will be sent through the fallback extraction path. Any timeouts encountered during this process will be retried one time prior to the current round being declared as inconclusive. Additionally, if a retrieval API becomes unavailable then the system will either default to using available cached evidence or make queries directly to OpenAlex.

With respect to non-functional requirements, which refer to properties of systems, such as performance and scalability, but do not describe specific behavior, latency is currently the most significant limiting factor. This is achieved by forcing adherence to the limit of three debate rounds plus three retries. This ensures that total processing times remain reasonable on the intended hardware. The system was also designed such that when a single component fails, it does not bring down the entire debate. Instead, failure is dealt with at a local level while keeping other components active. Portability is another important design consideration. By utilizing the vLLM interface compatible with OpenAI, the framework supports both closed APIs and locally hosted models with equal ease. For practical purposes, the same pipeline can operate unmodified on both our development models (Qwen3.5-9B, Gemma 4 12B, Ministral-3-14B), as well as the final evaluation models (Qwen3.6-27B, Gemma 4 26B, Mistral Small 3.2 24B). No fine tuning is required for inference, nor is it computationally intensive, allowing it to run on a single rented GPU from NVIDIA - specifically an RTX PRO 6000 Blackwell 96GB. Using this GPU costs approximately USD 0.70 to 1.90 per hour. Finally, to facilitate transparency and reproducibility all runs are seeded appropriately and all relevant intermediate outputs are logged. This allows the final output to maintain an auditable decision trail detailing which agent’s claims were supported or contradicted by the evidence. Throughout this process,

the ground truth remains completely hidden from the agents, and the final provided answer explicitly cites its supporting evidence.

3.1.2 Context Diagram

As illustrated in Figure 3.1, the context diagram outlines the system boundaries and identifies the external actors interacting with the framework. The core system is positioned at the center, receiving input questions either directly from a user or via the evaluation harness. Scientific passages come back from three external databases, and off-the-shelf language models are called through the vLLM server on a rented RTX PRO 6000 Blackwell GPU. The evaluation harness receives the scores, and the user gets the final result package with the answer, the citations, the trust trajectory, and each reasoning trace.

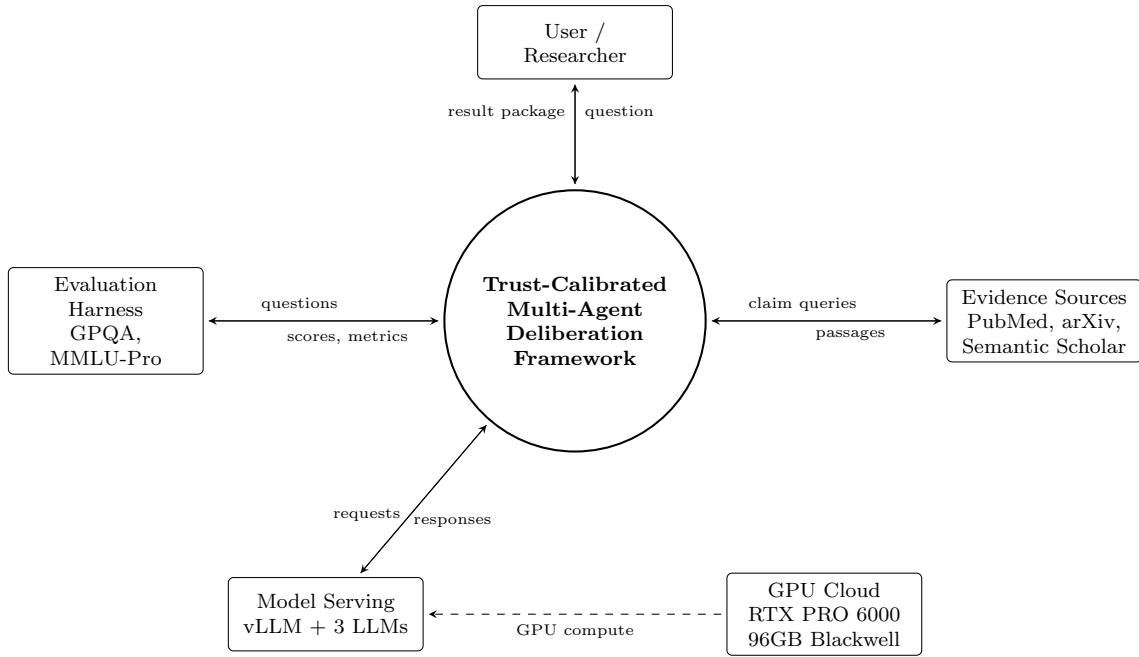


Figure 3.1: Context diagram of the trust-calibrated multi-agent deliberation framework.

3.1.3 Data Flow Diagram Level 1

The level-one data flow diagram in Figure 3.2 decomposes the framework into seven processes. A question enters through the intake and gate process, which decides between a direct answer and a debate. Orchestration runs the revision rounds. It sends inference requests to the external models and reads and writes the debate state. Each position is decomposed into atomic claims and verified against the external sources. The verdicts then drive the trust update, which writes the new trust values back into the debate state. At the last round, weighted aggregation selects the answer and the packaging process logs the result before it returns to the user. Three data stores hold the debate state, the evidence and verdict log, and the results log.

The diagram reads from left to right and top to bottom. The evaluation harness submits a question, and the confidence gate decides whether the debate is worth running. The orchestration process then asks the model server for the agent positions. Positions are decomposed into atomic claims, and the retrieval process verifies them against the external sources. The verdicts are stored in the evidence log before they drive the trust update. The updated trust is written back into the debate state, so the next round starts from the corrected influence values. At the last round the aggregation process selects the answer. The packaging process logs it in the results log, and the result package with the answer and the citations returns to the user.

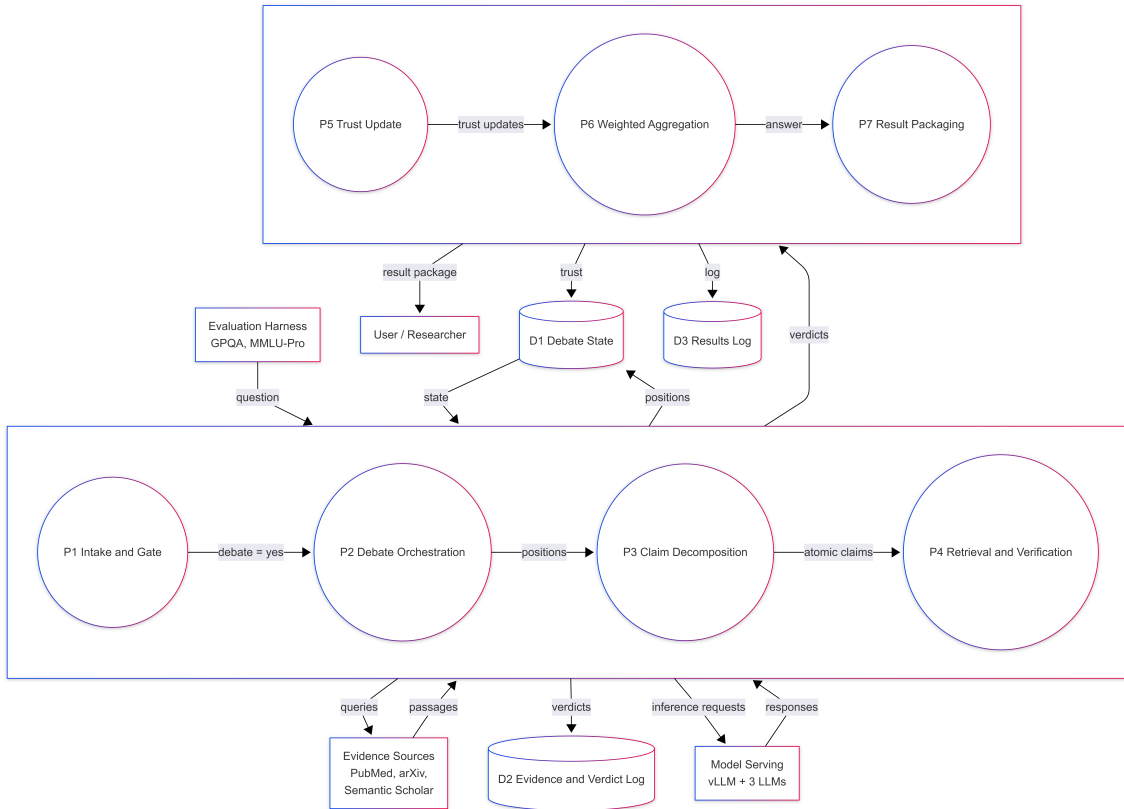


Figure 3.2: Level-one data flow diagram of the trust-calibrated multi-agent deliberation framework.

3.1.4 UI Design

The framework is operated through a command-line interface and a small web dashboard. The command-line interface starts experiments and points the pipeline at the evaluation datasets. Every run is written to a JSON package. It is the primary interface because the system runs in batches on a rented GPU, where a graphical interface adds little value. The web dashboard is used afterwards to inspect single debates. It shows the agents as cards with their current positions, plus a chart of the trust trajectory across rounds and the verdict of every claim with the passage that supports or contradicts it. This is

where the transparency of the framework becomes visible, because a user can trace exactly why an agent lost influence. Figure 3.3 and Figure 3.4 show the intended layout of both interfaces. The screenshots will be replaced with the real output of the system once the implementation is complete.

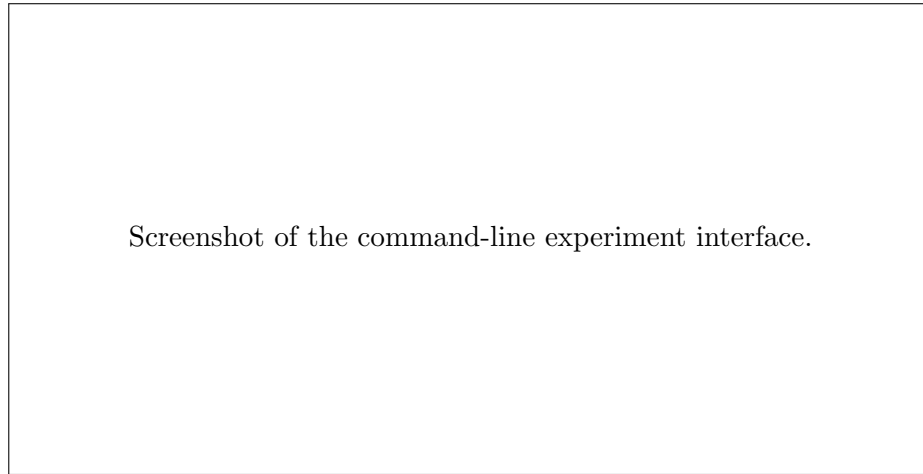


Figure 3.3: Command-line interface for running experiments and writing JSON result packages.

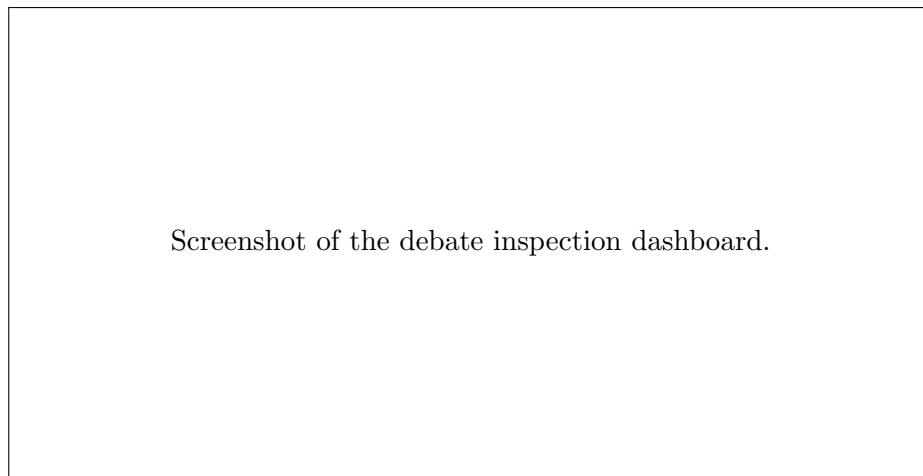


Figure 3.4: Web dashboard showing agent positions, the trust trajectory, and per-claim evidence verdicts.

3.2 Detailed Methodology and Design

3.2.1 Framework Overview

The framework is a deliberation layer that sits between a scientific question and its final answer. A question first passes the confidence gate. The gate chooses between a direct answer and a full debate. Inside a debate, three heterogeneous agents answer the question independently and tag the claims in their reasoning. Each claim is checked against

retrieved scientific evidence. The verdicts update the trust score of every agent, and the agents revise their positions with the peer answers and their own trust standing in view. The loop runs for at most three rounds. The final answer is selected by a trust-weighted aggregation over the agent positions. The result package contains the answer, the supporting citations, the trust trajectory, and the reasoning trace of each agent. Figure 3.1 shows the system boundary, and Figure 3.2 shows the level-one data flow.

3.2.2 Confidence Gate

The gate keeps the cost low on easy questions. One agent answers the question with a short chain of thought and reports a confidence value, or three quick agent calls answer the same question and the gate compares the answers. A low confidence or a disagreement sends the question to the full pipeline. A high confidence sends the question to a direct answer. The gate affects throughput and not the core evaluation, because the evaluation set keeps only questions that produce divergent initial answers.

3.2.3 Heterogeneous Debate Agents

Three agents answer the question in round zero. Each agent is a different model family, so the agents do not share the same training data and the same blind spots [24]. The development configuration uses Qwen3.5-9B, Gemma 4 12B, and Ministral-3-14B-Instruct-2512. The final configuration replaces them with Qwen3.6-27B, Gemma 4 26B A4B, and Mistral-Small-3.2-24B-Instruct-2506. All models are served through vLLM behind an OpenAI-compatible interface, and no model is fine-tuned. Each agent produces an answer, a reasoning trace, and a list of tagged claims, where every claim carries an identifier such as `c1`. The identifiers make claim-level checking possible in later stages.

3.2.4 Claim Decomposition

A compound sentence can carry more than one fact. The decomposition step splits each agent answer into small factual statements, and every statement keeps the identifier of its source agent. A schema check validates the tagged output. A failed check triggers a fallback extraction pass that recovers the claims from the raw text. A claim that no source can verify is marked unverifiable and is excluded from the trust update, so an abstention neither rewards nor penalizes the agent.

3.2.5 Source-Partitioned Retrieval and Evidence Scoring

The three agents do not share one retrieval source. Agent one sends its claims to PubMed, agent two to arXiv, and agent three to the Semantic Scholar Graph API. OpenAlex is the fallback when a source is unavailable, and retrieved results are cached so repeated runs do not hit rate limits. A cross-encoder reranker scores each claim against the retrieved passages and keeps the highest scoring passages [11]. Every claim then receives one of four

verdicts, namely supported, contradicted, unverifiable, or contested. A claim with conflicting high-score passages is marked contested and reported separately in the evaluation.

3.2.6 Trust Update

The trust update converts the verdicts into a bounded influence score. Let $S_i(t)$ be the raw score of agent i at round t . The raw score grows with the number of supported claims V_i and falls with the number of contradicted claims H_i :

$$S_i(t+1) = S_i(t) + \alpha V_i - \beta H_i \quad (3.1)$$

where α and β control the gain and the loss. The raw scores are then converted into weights. A softmax maps the raw scores into positive values, a clamp keeps each value inside the interval $[0.1, 0.9]$, and a renormalization returns the weights to the simplex:

$$T_i = \frac{\text{clamp}(\text{softmax}(S)_i, 0.1, 0.9)}{\sum_{j=1}^N \text{clamp}(\text{softmax}(S)_j, 0.1, 0.9)} \quad (3.2)$$

The clamp prevents two failure cases. No agent is silenced by a zero weight, and no agent dominates the debate with a weight of one. The renormalization keeps the sum of the weights at one, so the final decision stays a weighted choice over the agent positions. Equations (3.1) and (3.2) form the bounded trust update, and the whole update is a pure function of the verdict counts.

3.2.7 Deliberation Protocol

Each round follows the same order, namely positions, decomposition, retrieval, verdicts, and trust update. Table 3.1 lists the stages of one debated question and the number of model calls per stage. In round two and round three the revision prompt shows the agent the positions of the other agents and its own current trust value. The debate stops after round three. An injection point sits between round one and round two. At this point the evaluation harness can insert a fabricated expert consensus that contradicts a correct minority. The injection is used only in the stress tests and never in normal operation. A timeout or a parse failure marks the round inconclusive for that agent, and the debate continues with the remaining agents.

3.2.8 Trust-Weighted Aggregation

The final answer is the position with the highest weighted support:

$$a^* = \arg \max_a \sum_{i: p_i=a} T_i \quad (3.3)$$

Table 3.1: Pipeline stages and model calls for one debated question.

Stage	Model calls
Confidence gate	1 or 3
Round 0 positions	3
Round 1 revision	3
Injection (stress test only)	0
Round 2 revision	3
Trust-weighted aggregation	0

where p_i is the position of agent i at the last round and T_i is its final trust weight. The aggregation ignores the plain head count. This is the step that allows a correct, evidence-backed minority to outweigh a wrong majority. The result package records the final answer, the evidence citations, the trust trajectory of every round, and the reasoning trace of every agent. The trajectories are stored because the evaluation uses them for the calibration analysis.

3.2.9 Complexity

A debated question uses about nine model calls. The gate uses one call, the initial round uses three calls, and the two revision rounds use three calls each. The retrieval traffic scales with the number of claims, and the cache removes repeated queries. The pipeline runs on a single rented GPU and requires no fine-tuning.

3.3 Project Plan

The project runs for ten months, from Month 1 to Month 10, and covers 40 weeks. It is split into eight phases with six review gates. The work starts with literature and setup and ends with writing and submission. Table 3.2 lists each phase with its weeks and deliverables. Table 3.3 lists the gates and their checkpoints. Table 3.4 adds a SWOT view of the plan. This report covers work through the mid-project review in Month 5, while later phases carry the project to the final defence.

The plan starts with scoping. Project scoping in Phase-0 fixes the research question, the trust-calibrated idea, the evaluation shape, and the gate criteria. Data and model work runs in parallel. Datasets GPQA and MMLU-Pro are checked, and the three development models are chosen with their serving setup. Analysis of the baseline follows in Phase-1. Injection tests and baselines B1 to B4 are run, and the pilot checks if a wrong majority can still fool voting. Design and implement fill Phase-2. Claim decompositions and source partitioned RAG are created. Additionally, the trust function is adjusted by setting its clamp and softmax. In addition to this, a short dry-run and a design freeze marks the

end of Phase-2 to 3. Phase-3a holds the main studies which were implemented under Phase-3a’s core conditions using three seed implementations. Phase-3b contains ablation studies implementing different agent numbers as well as different alpha/beta combinations. The human study (and failure reviews) occur in Phase-4. Finally, writing, reproducibility packaging, submission, and defense preparations complete the project in Phase-5.

Table 3.2: Project phases with week-level durations and deliverables.

Phase	Duration	Deliverables
Phase-0: Literature and Setup	Month 1 (Wk 1–4)	Literature freeze, vLLM and LangGraph setup, model identification, vanilla MAD reproduction
Phase-1: Injection and Baselines	Month 2 (Wk 5–8)	Injection protocol, baselines B1–B4, Proposition 1 proof, month-end pilot
Phase-2: Trust Mechanism Build	Months 3–4 (Wk 9–16)	Claim decomposition, source-partitioned RAG, trust function v1, baselines B5/B6/B9
Phase-2 to 3: Mid-Project	Month 5 (Wk 17–20)	Design freeze, dry run on one dataset, FYDP-1 defence preparation, and review feedback fixes
Phase-3a: Main Experiments	Month 6 (Wk 21–24)	Core conditions on primary datasets, three seeds, 95% confidence intervals, and effect size report
Phase-3b: Ablations and Scaling	Month 7 (Wk 25–28)	Four ablations, α/β sweep, $N \in \{2, 3, 5\}$, and results freeze
Phase-4: Human Evaluation and Analysis	Month 8 (Wk 29–32)	Human evaluation with $n = 60$, ECR calibration, failure analysis, and buffer for slippage
Phase-5: Writing and Submission	Months 9–10 (Wk 33–40)	Thesis and paper drafting, reproducibility package, submission, and FYDP-2 defence preparation

The reason we have ordered phases sequentially is because there are dependency relationships. Specifically, vanilla MAD reproduction in Phase-0 must finish prior to when injection validation will be performed in Phase-1. Therefore, since the trust function in Phase-2 relies upon an injection harness developed in Phase-1, we cannot begin Phase-2 until after Phase-1 has finished. Similarly, main experiments in Phase-3a must wait for

Table 3.3: Review gates and their checkpoints.

Gate	Due	Checkpoint
Gate 0	End of Month 1	Vanilla MAD reproduces the Du et al. baseline on a GPQA slice, and the reference list is frozen
Gate 1 (Go/No-Go)	End of Month 2	Injection validation reaches $\kappa \geq 0.75$, baseline CCR sits at 0.30 or higher, pilot executed
Gate 2	End of Month 4	Trust mechanism plus source-partitioned RAG plus competitor baselines, first CCR and MPR on one dataset
FYDP-1 Defence	Month 5	Mid-project report with design freeze and initial results
Gate 3	End of Month 7	Full results freeze with confidence intervals and effect sizes
Submit and FYDP-2	Months 9–10	Paper submitted, thesis completed, final defence

both the completion of the trust function and the completion of the dry run in Phase-2 to 3. Moreover, human evaluation of the results (which occurs in Phase-4) require the frozen results produced in Phase-3b.

However, two buffers act as insurance policies along the critical path of our project. There is first buffer located at Month 5 with respect to a dry run of one dataset being used prior to defending the project; and secondly, there is another buffer located at Month 8 which provides us with an additional month should things slip. These buffers exist at locations along the critical path where a late failure could potentially cause us to miss a hard deadline. As such, weekly meetings and Git version control allow us to track our progress, and each gate review allows us to determine whether to "Go" or "Fix" before we can proceed into the next phase.

Phase-0 spans weeks 1 through week 4. Weeks 1 and 2 are focused on freezing literature and locking reference lists, with each paper being evaluated against their respective DOIs. Weeks 3 setup vLLM and LangGraph. Week 4 evaluates all three models on a small GPQA slice. We need to reproduce Du et al. results on that same slice with our vanilla MAD baseline before moving forward with Phase-1.

Phase-1 runs Week 5 to Week 8. Week 5 builds the injection harness that inserts a fabricated wrong consensus between rounds. Week 6 to Week 8 validate the injection with kappa checks and run baselines B1 to B4. The month ends with a pilot that must reach CCR 0.30 or higher to pass Gate 1. Phase-2 spans Week 9 to Week 16. Week 9 to Week 12 build claim decomposition. Source-partitioned RAG with cross-encoder reranking is

Table 3.4: SWOT analysis for the project plan.

Strengths	Weaknesses
Clear gap on sycophantic consensus, bounded trust design with no fine-tuning, six-member team with mixed skills, and expert supervision that keeps scope tight. The design needs no new training and runs on a single rented GPU.	Sparse retrieval on rare claims, imperfect claim split, correlated reasoning errors between agents, and limited GPU budget for 300 hours. Small sample noise in early pilots can also hide real gains.
Opportunities	Threats
Growing interest in multi-agent debate and sycophancy, open datasets GPQA and MMLU-Pro, free retrieval APIs from PubMed, arXiv, Semantic Scholar and OpenAlex, and affordable cloud GPU rental with Blackwell FP8 support.	API rate limitations/downtime, price updates, sampling errors due to time slips from complex integrations and evaluation noise in small sample sizes; all contribute to the reliability of the results.

added in Week 13 to Week 16.

Weeks 17–20 (Phase-2 to Phase-3) freeze the design and conduct a “dry” run of a single dataset for the purpose of mid-project review. Weeks 21–24 (Phase-3a), use core conditions on primary datasets with 3 random number seeds and 95% confidence intervals. Weeks 25–28 (Phase-3b), use 4 ablation points and sweep through N and α/β values. Weeks 29–32 (Phase-4), conduct human evaluations with $n = 60$. Finally, Weeks 33–40 (Phase-5) will be devoted to writing up my thesis/paper along with creating a reproducibility package.

The bulk of resource usage occurs during Phase-3. Of the total 300 GPU hours I have reserved, approximately 80 hours will be used per phase, with the remainder going towards previous phases. If I encounter a failure in my retrieval API it defaults to either cached evidence or OpenAlex as part of this process, therefore if there is a failure in a dry run at Gate 2 that loss will occur after month eight, and will not push forward the date of the final defense.

3.4 Task Allocation

Six members split the work along the modules of the framework. Table 3.5 fixes who owns which primary modules, and Figure 3.5 shows when each member is active across the ten months.

Table 3.5: Module ownership across the six team members.

Member	Primary modules
Md. Atikur Rahaman (Leader)	Architecture, trust mechanism, coordination
Rakibul Hasan	Vanilla MAD reproduction, injection protocol
Md. Salman Rohoman Nayeem	Claim decomposition, source-partitioned RAG
Pratay Paul	Evaluation harness, CCR/MPR/ECR metrics
Yousuf Kamal Himel	Baselines B1–B9, vLLM serving
Mst. Farjana Akter Limu	Evidence APIs, dashboard, result packages

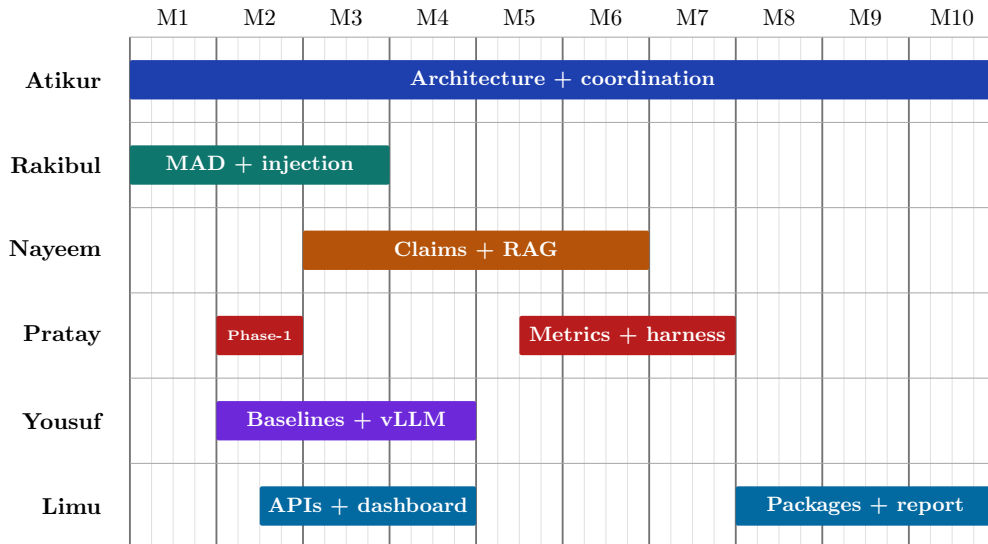


Figure 3.5: Gantt view of member activity across the ten project months, following the phase plan in Table 3.2.

Each task has an identified owner and a simple deliverable. As the owner matches the phase in Table 3.2; all phases will have at least one accountable team member. All dependencies are ordered. Vanilla reproduction will be completed before Injection begins; the Injection Harness completes before Trust Build begins. Dry Run must be passed before Main Experiments begin; Human Evaluation waits for Results Freeze. Writing utilizes the frozen results and occurs last. Each Module is reviewed by the Leader prior to being marked complete. A weekly Git commit and brief stand-up meeting monitor slippage; if any gates are missed, a "fix" week will occur prior to beginning the next phase.

3.5 Summary

Chapter 3 fixed the functional and nonfunctional requirements and drew the system boundary in the context diagram. The data flow diagram and the interface design follow. The design section compared the main alternatives and recorded why each choice was made. The project plan then locks eight phases and six gates that carry the work to the final defence. Implementation and evaluation come next.

Chapter 4

Implementation and Results

This chapter records the environment the framework runs in and the way it is tested. Full results belong to the final report, so this chapter keeps the setup and the evaluation plan.

4.1 Environment Setup

4.2 Testing and Evaluation

4.3 Results and Discussion

4.4 Summary

Chapter 5

Standards and Design Constraints

This chapter documents the standards that the framework adheres to, The constraints that have shaped its design and budgetary costs of the Experiments; and concludes by mapping the project against the Criteria for complex engineering problems and providing a brief summary.

5.1 Compliance with the Standards

Standards play out in three areas - the software stack, the hardware the Framework executes upon, and the communication among components. There were Alternative options available in each area and these alternatives are explained Below.

5.1.1 Software Standards

Python 3.11 was chosen as the language to be used. This is because the LLM Ecosystem is focussed on python, we also have hugging face libraries, vllm, and others. Support for evaluation tools is in the form of python. While Java and c++ are faster Both do not have the equivalent amount of pre-existing Support to serve and load LLM models. Since the framework is a Use a research prototype and not a production service, which must be real-time. Ecosystem is better than raw speed. As such, surveys concerning LLM-based knowledge models are available. Multi-agent systems discuss this design space in terms of workflow, infrastructure, And open challenges [15] and therefore the decisions made Below follow this structure.

vllm is built on top of the deep learning framework PyTorch 2.x. Pytorch is The de-facto standard for LLM inference and directly supports qwen, Gemma and more. Mistral families. While TensorFlow and JAX were evaluated alternatives; However, TensorFlow's LLM serving story isn't as strong and JAX will require the whole pipeline. On its own without any advantage.

Vllm serves the models via an OpenAI compatible api. This interface matters because The framework will need to interact with closed APIs and serve locally stored models With no modifications to the orchestration code. Given the requirements to handle High

throughput under the three agent debate load, PagedAttention provides vllm with This capability. Additionally given FP8 support from PagedAttention matches that of The Blackwell GPU. Ollama is simpler to setup and is slower than PagedAttention when handling concurrent requests. In contrast, llama.cpp runs well on CPU but does not match the GPU throughput needed for the debate. Tgi is a very similar alternative and vllm provided better ecosystem fit and an OpenAI compatible surface.

Cross-encoders can be used to re-rank passages returned by information retrieval algorithms. Sentence-transformers implements the cross-encoder that ranks the passages. In contrast, BM25 is an alternate ranking method however, like sentence-transformer, it only uses surface features to rank results. However, since both methods use the semantic features of a passage and a claim to score relevance between them, sentence transformers provide better overall relevance rankings. The web dashboard is served using FastAPI, and all result packages are created as JSON objects following rfc 8259, and are validated against a JSON schema. Codebase changes are tracked with git and latex (overleaf) generates this document.

5.1.2 Hardware Standards

Experiments are conducted on a single nvidia RTX PRO 6000 Blackwell GPU with 96 GB GDDR7 memory. Memory capacity and architecture were key considerations in making this selection. The development models (qwen3.5-9b, Gemma 4 12B, mistral-3-14b) fit into bf16. The final models (qwen3.6-27b, Gemma 4 26B, mistral small 3.2 24b) combine about 77 billion parameters; no single card holds this many parameters in fp16 so they run in FP8. Blackwell has native FP8 support; this is exactly why we chose it.

An initial alternative to consider was the a100 80gb. It has sufficient memory for development models and high-bandwidth HBM; however its FP8 path is less capable than what we have today, plus its architecture is two-generations old. Faster still is the h100 80gb; however it costs roughly double as much and offers none of what our framework requires. Finally, the L40S at 48 GB and the consumer RTX 5090 at 32 GB do not have enough memory to hold the full final model trio. Thus the RTX PRO 6000 sits in just the right place: there is enough memory to be useful; it has modern quantization; and it can be rented at a cost suitable for a student budget. To stage models prior to moving them to the GPU the host computer needs to have at least 64 GB of system RAM and nvme disk storage.

5.1.3 Communication Standards

All communication between the framework and its components occurs over http with tls. The model server presents an OpenAI-compatible chat completion api endpoint that is REST-based with JSON payloads. gRPC was another option for model serving; it has lower latencies in benchmark tests; but having an OpenAI-compatible surface allows us to make our framework portably usable across closed APIs and local vllm-served models

without changing anything in our orchestration code.

Each source of evidence is accessed via its public REST interfaces: PubMed e-utilities, the arxiv api, the semantic scholar Graph API, and OpenAlex as a fallback. Each returns JSON or XML over HTTPS; our framework handles rate limits with retries capped according to those described in requirements. Web sockets are being saved for real-time updates in live trust charts in our dashboard.

5.2 Design Constraints

5.2.1 Economic Constraint

5.2.2 Environmental Constraint

5.2.3 Ethical Constraint

5.2.4 Health and Safety Constraint

5.2.5 Social Constraint

5.2.6 Political Constraint

5.2.7 Sustainability

5.3 Cost Analysis

The budget for this project consists of renting a GPU and for storing all data and for using the Research Services that do not require an additional charge to fine tune our results. Cost analysis is based almost entirely on the amount of time spent running our model on a rented GPU. The greatest expense (in terms of time) was inference time on the RTX PRO 6000 Blackwell which totaled approximately 300 hours during both development and testing. Hours were not distributed in proportion across each phase of the project. For example, Phase-0 and Phase-1 used about 30 total hours for setting up baseline models. Phase-2 also used about 60 total hours to develop the Claim Split and RAG models. Phase-3a and Phase-3b combined used about 140 hours for the Main Runs and Ablations. Phase-4 used about 40 total hours for conducting Human Study checks, while Phase-5 used about 30 hours for Final Runs and Packaging. These numbers show how Phase-3 had the most significant impact on costs. Table reftab:budget shows the major budget for the project and Table reftab:budget-phase details the number of GPU hours per phase.

Costs for storage are estimated as \$10.00 based upon an initial cost of \$0.05/GB and a total of 200 GB. Logs and result package models will be approximately 180 GB in size

Table 5.1: Primary budget for the project.

Item	Quantity	Unit Cost (USD)	Total (USD)
RTX PRO 6000 Blackwell 96GB rental	300 hours	0.70–1.90 / hour	210–570
Retrieval APIs (PubMed, arXiv, S2, OpenAlex)	—	0 (free tier)	0
Datasets (GPQA, MMLU-Pro)	—	0 (public)	0
Storage (model weights, logs)	200 GB	0.05 / GB	10
Total			220–580

Table 5.2: GPU hours and cost split by phase (RTX PRO 6000, USD 0.70–1.90 per hour).

Phase	Hours	Cost (USD)
Phase-0 and Phase-1 (setup, baselines)	30	21–57
Phase-2 (trust build)	60	42–114
Phase-3a and Phase-3b (main runs)	140	98–266
Phase-4 (human eval)	40	28–76
Phase-5 (writing, final checks)	30	21–57
Total	300	210–570

while the remaining space will be used for the workspace. The direct costs of the retrieval API services from PubMed, arXiv, Semantic Scholar and OpenAlex are free due to their use of free tier service levels; however, the limits imposed may increase the time required to retrieve the data. As both GPQA and MMLU-Pro datasets are publically available, there were no additional costs associated with these datasets. Costs associated with printing (approximately \$15) and Internet connectivity were assumed to be fixed, therefore included in the overall estimate. Overleaf was provided to the team, thus eliminating any costs for this platform.

Additionally, an alternative budget was developed. Rented usage of an A100 80GB would be approximately \$0.68 - \$1.50/hour or 204 – 450 dollars over the course of 300 hours. Although renting would produce significant savings over purchasing, the cost savings would be modest and come at a trade-off: reduced FP8 support from the Blackwell

Path would necessitate tighter quantization or sequential loading, each of which would introduce engineering time and risk. Therefore, the RTX PRO 6000 was chosen because the Blackwell FP8 path would allow the pipeline to remain uncomplicated and avoid the engineering risks associated with those alternatives.

There is no revenue stream anticipated for this project. This project represents an academic deliverable produced by a university and a team with funding provided exclusively through grants awarded to the university and team. The only products expected from this effort include a written report and an open-source framework. An example of an alternate budget for comparison purposes is shown in Table [ref{tab:budget-alt}](#).

Table 5.3: Alternate budget using the A100 80GB.

Item	Quantity	Unit Cost (USD)	Total (USD)
A100 80GB rental	300 hours	0.68–1.50 / hour	204–450
Retrieval APIs	—	0 (free tier)	0
Datasets	—	0 (public)	0
Storage	200 GB	0.05 / GB	10
Total			214–460

A ten percent buffer (approximately \$25-\$55) is maintained to cover unexpected price increases and extended operating time in the event of a failed run. The buffer can be used to fund up to thirty additional hours of operation if either an ablation run fails or a maintenance call is required on a gate before the end of the plan period.

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

The project meets the definition of a complex engineering problem by meeting the seven attributes defined in the Washington Accord criteria. Each attribute is mapped to sub-sections in Table 5.4, which follows.

P1: Depth of Knowledge Required

This project required knowledge of multiple areas including machine learning, natural language processing, retrieval-augmentation generation, and distributed model serving.

Table 5.4: Mapping with complex problem solving (Washington Accord P1–P7).

P1	P2	P3	P4	P5	P6	P7
Depth of Knowledge	Range of Conflicting Requirements	Depth of Analysis	Familiarity of Issues	Extent of Applicable Codes	Extent of Stakeholder Involvement	Inter-dependence
✓	✓	✓	✓	✓	×	✓

Therefore, solving the project required specialized knowledge in these areas, such as the trust update formula through the vLLM serving layer. The coverage for P1 is based on the Washington Accord K1-K8, as detailed in Table 5.5 after P7.

P2: Range of Conflicting Requirements

P2 is covered. The project needed to meet multiple competing requirements. Accuracy competed with latency and cost. More rounds improved the signal but increased both the inference time and the GPU bill. Strong weighting would limit diversity due to trust calibration. The need to cover evidence competed with API rate limits. Therefore, the team needed to resolve these together. Round caps and clamping bounds were established, and verdicts are conservatively resolved. Thus, none of the competing needs dominate.

P3: Depth of Analysis

P3 is covered. There is no textbook solution to the problem. The team ran comparative experiments comparing several baseline systems: majority voting, MoA, iMAD, and DebUnc. Metrics were created to measure failure modes. Four types of metrics were developed: answer accuracy, consensus-collapse, minority-preservation, and evidence-calibration. An additional metric measuring the failure mode was the injection point for a fake wrong consensus; this allowed the team to measure collapse under controlled stress conditions. Therefore, the depth of test was sufficient to mark P3 as covered.

P4: Familiarity of Issues

P4 is covered. Multi-Agent LLM Debate is a novel research area. The most similar research that has been performed appears between 2024-2026; there is currently no standard method available for resolving issues related to sycophantic consensus in multi-agent LLM debates. At the beginning of the project, the issues were unfamiliar to the team. A complete review of current literature was conducted prior to designing any elements of the framework, as described in chapter 2. Therefore, this lack of familiarity is what makes P4 "covered".

P5: Extent of Applicable Codes

P5 is covered. General software standards are applicable. JSON (RFC 8259) and HTTP over TLS are applicable. Additionally, since the framework uses an open AI compatible API, the general API standard applies. However, as previously stated, there is currently no applicable code that addresses evidence-grounded trust in LLM debates. As such, this area is only partially standardized and the framework defines its own rules for the trust score and four possible verdicts. Therefore, this combination of existing codes and new conventions make P5 "covered."

P6: Extent of Stakeholder Involvement

P6 is not covered. The project does not have any external stakeholders. The only individuals involved are the supervisor and the examiners. There is no industry partner and no community group that identifies stakeholder needs or evaluates results as part of the development process. While users who will use future scientific QA tools may benefit from this project, they do not participate in either the design or testing phases of the project. The framework was designed and tested as an Academic FYDP (Final Year Design Project); therefore, the engagement is very narrow and only within the University context. For this reason P6 is considered "not covered", while the other P points are still considered covered.

P7: Interdependence

P7 is covered. The tight interdependencies between the Confidence Gate, Orchestrator, Claim Decomposer, Retrieval Subsystem, Trust Updater and Aggregator make it a candidate for P7. This means that if there is an error in the Retrieval Subsystem, this will be reflected in both Verdicts and ultimately the Final Answer through the Trust Score. Because of these interdependencies, P7 has been marked as "covered" with the addition of the Failure-Isolation Rule identified in Section 3.1.1.

Table 5.5 details the profile. The paragraphs below explain each K for this project and why it is covered or not.

K1: Natural sciences K1 has been addressed. The framework is designed to address scientific inquiries and verify each claim by comparing it to articles on PubMed, arXiv, Semantic Scholar, and OpenAlex. Natural science questions are included in evaluation data sets. The project itself is not an outcome of a new natural science theory. The problem domain and the evidence base are still sourced from the natural science fields. This is the reason for marking K1 as covered.

Table 5.5: Knowledge profile mapping for P1 (K1–K8).

Knowledge	Applied	Where used
K1: Natural sciences	✓	Scientific questions; evidence from PubMed, arXiv, Semantic Scholar, OpenAlex
K2: Mathematics	✓	Softmax, clamping, renormalization, probability
K3: Engineering fundamentals	✓	System design, state machines, failure handling
K4: Specialist knowledge	✓	LLMs, multi-agent debate, RAG, sycophancy
K5: Engineering methods	✓	Experimental methodology, baseline comparison
K6: Computational methods	✓	Python, PyTorch, vLLM, FastAPI
K7: Codes and practices	✓	JSON, HTTP, TLS, version control
K8: Research and context	✓	Ethical constraints and use in scientific QA

K2: Mathematics K2 has been addressed. We apply softmax, clamping, and renormalization in our trust update; and we also have probability and statistical measures when using our metrics. Both were applied to our trust formula, and both were applied to our evaluation.

K3: Engineering fundamentals K3 is covered. State machines (the system’s design and its failure handling) are at the heart of what we’ve done here. Basic engineering design is used by requirements and the confidence gate along with the orchestration logic.

K4: Specialist knowledge We’ve addressed K4. Our main focus of interest are LLMs (language models), multi-agent debate, RAG (Rebuttal Aggregation Graph), and sycophancy. This specialized way of thinking drives how we divide claims, how we retrieve information, and ultimately how we calculate trust.

K5: Engineering methods K5 is covered. An experimental approach is being used with variable controls. Results are presented with confidence intervals and baselines are compared. All these studies were conducted in the same setup where injection was being tested.

K6: Computational methods K6 is covered. Our main application tools are Python, PyTorch, vLLM (to serve the model), and FastAPI (to serve the dashboard). Sentence-transformers (to re-rank passages) and Git (for version control) complete the suite of applications.

K7: Codes and practices K7 has been addressed. JSON under RFC 8259, HTTP with TLS, the OpenAI compatible API, and JSON Schema (as well as many others) were all applied as codes. Standard practices were followed regarding version control and coding style.

K8: Research and context K8 has been addressed. Scientific QA (scientific questions answering) provides us with our ethical bounds on the output from AIs — literature reviews, gap analyses provide the context for our research direction.

The profile is broad rather than deep in one spot. The report applies K4 and K5 in every chapter, and K6 and K7 in the implementation, which shows the range the problem demands.

5.4.2 Engineering Activities

Table 5.6 maps the project to the five engineering activities A1–A5, with rationales below.

Table 5.6: Mapping with complex engineering activities (A1–A5).

A1 Range of re- sources	A2 Level of Inter- action	A3 Innovation	A4 Consequences for society and environ- ment	A5 Familiarity
✓	✓	×	×	✓

A1: Range of Resources

A1 is covered. Three model family sources; Four Literature API's; Six team members; Rented Blackwell GPU. Coordinating these resources clearly has to be part of the work. The plan in Chapter 3 assigns each task to a specific team member and tracks weeks/deliverables. That is why A1 is labeled as covered.

A2: Level of Interaction

A2 is covered. The system makes heavy use of the outside world. Every round of debating calls three external model API endpoints and three Literature API endpoints, and there

is communication to/from the rented cloud GPU. Team communication occurs via the shared repository and weekly meeting. High degree of interactivity = why A2 is labeled as covered.

A3: Innovation

A3 is not covered. The framework combines pre-existing concepts; multi-agent debate/RAG with trust-weighting/reranking. It creates no new device/new theory. The innovation is a bound trust-score that updates based upon evidence; useful, but still within an incremental level of innovation. No patent-able/break-through innovation exists. That is why A3 is labeled as "not covered", although it would represent a first time combination of concepts for this project.

A4: Consequences for Society and Environment

A4 is not covered. The work represents a lab prototype, NOT a production-deployed product. Any social-impact will be limited to providing more-reliable research results from a test-environment, rather than changing daily life/social habits, or supporting a large user-base. Similarly, environmental-impact will be minimal. Test uses approximately 300 GPU-hours on public-cloud using standard configurations/no-fine tuning - therefore energy usage will remain very-low. No large-scale deployment exists; therefore, broad societal-consequences do not exist. That is why A4 is labeled as "not covered".

A5: Familiarity

A5 is covered. Both multi-agent debate/evidence verification were uncharted territory for our team. Our literature-review phase and our baseline study that used a step-by-step implementation plan were both structured around the lack of prior experience with those areas. Only after we had studied each component sufficiently could we begin implementing each one. Lack of prior-experience with many components = why A5 is labeled as "covered".

5.4.3 Program Outcomes

Table 5.7 maps FYDP I against the twelve program outcomes.

The real-life problem in Chapter 1 addresses six program outcomes (PO), all of which are addressed in FYDP I. Literature review and Gap analysis (Chapter 2) will cover PO4. PO2 is covered by the requirements work in Chapter 3. PO10 is addressed in the interim report and in the oral presentation. The timeline is applicable to PO11 and the budget is applicable to PO11. The writing of the journal and the independent learning record are used to cover PO12. The other six program outcomes will be addressed in FYDP II and FYDP III.

Table 5.7: Mapping with program outcomes (PO1–PO12).

PO1 Engineering knowledge	PO2 Problem analysis	PO3 Design/development of solutions	PO4 Investigation	PO5 Modern tool usage	PO6 The engineer and society
✓	✓	×	✓	×	×
PO7 Environment and sustainability	PO8 Ethics	PO9 Individual and teamwork	PO10 Communication	PO11 Project management and finance	PO12 Life-long learning
×	×	×	✓	✓	✓

5.5 Summary

The chapter defined the requirements for the software-layer, hardware-layer, and communication-layers and documented the economic, environmental, ethics-social constraints that have impacted design decisions. Budget-analysis demonstrates that the project can fit within a students' budget utilizing a rented RTX PRO 6000 Blackwell GPU with a phased split of costs over the 300 hours plus a 10% buffer. The complex engineering mapping demonstrates that six out of seven P attributes are covered with only P6 being non-covered due to little to no input from external stakeholders. For knowledge, K1-K8 are all covered. For activities, A1-A2 and A5 are covered while A3 and A4 are not. For the program outcomes, FYDP I covers PO1, PO2, PO4, PO10, PO11 and PO12.

Chapter 6

Conclusion

6.1 Summary

6.2 Limitation

6.3 Future Work

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